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# LABTRACK: A Console-Based Computer Laboratory Inventory System
# ITE 260 - Computer Programming 1
# Group 7

inventory = []


def get_non_empty(prompt):
    """Ask the user for a value that cannot be empty."""
    while True:
        value = input(prompt).strip()

        if value != "":
            return value

        print("Input cannot be empty.")


def get_positive_integer(prompt):
    """Ask the user for a positive whole number."""
    while True:
        try:
            value = int(input(prompt))

            if value > 0:
                return value

            print("Please enter a number greater than 0.")

        except ValueError:
            print("Invalid input. Please enter a whole number.")


def find_item(item_id):
    """Search for an inventory item using its Item ID."""
    for item in inventory:
        if item[0].lower() == item_id.lower():
            return item

    return None


def add_item():
    """Add a new inventory item."""
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print("\n========== ADD INVENTORY ITEM ==========")

item_id = get_non_empty("Item ID: ")

if find_item(item_id) is not None:
    print("Item ID already exists.")
    return

name = get_non_empty("Item name: ")
category = get_non_empty("Category: ")
quantity = get_positive_integer("Quantity: ")

while True:
    condition = get_non_empty(
        "Condition (Good/Damaged/Missing): "
    ).capitalize()

    if condition in ["Good", "Damaged", "Missing"]:
        break

    print("Please enter Good, Damaged, or Missing.")

location = get_non_empty("Location: ")

if condition == "Good":
    status = "Available"
elif condition == "Damaged":
    status = "Damaged"
else:
    status = "Missing"

item = [
    item_id,
    name,
    category,
    quantity,
    condition,
    location,
    status
]

inventory.append(item)

print("\nItem added successfully.")
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def display_items(records=None):
    """Display inventory records."""
    if records is None:
        records = inventory

    if len(records) == 0:
        print("\nNo inventory records found.")
        return

    print("\n================ INVENTORY LIST ================")

    print(
        f"{'ID':<10}"
        f"{'Name':<22}"
        f"{'Category':<18}"
        f"{'Qty':<6}"
        f"{'Condition':<14}"
        f"{'Location':<15}"
        f"{'Status':<12}"
    )

    print("-" * 97)

    for item in records:
        print(
            f"{item[0]:<10}"
            f"{item[1][:20]:<22}"
            f"{item[2][:16]:<18}"
            f"{item[3]:<6}"
            f"{item[4]:<14}"
            f"{item[5][:13]:<15}"
            f"{item[6]:<12}"
        )

    print("-" * 97)


def search_item():
    """Search inventory using an ID, name, category, or location."""
    if len(inventory) == 0:
        print("\nNo inventory records found.")
        return
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    keyword = get_non_empty("\nEnter search keyword: ").lower()

    results = []

    for item in inventory:
        if (
            keyword in item[0].lower()
            or keyword in item[1].lower()
            or keyword in item[2].lower()
            or keyword in item[5].lower()
        ):
            results.append(item)

    if len(results) > 0:
        print("\nSearch results:")
        display_items(results)
    else:
        print("\nNo matching inventory item found.")


def update_item():
    """Update an existing inventory item."""
    if len(inventory) == 0:
        print("\nNo inventory records to update.")
        return

    display_items()

    item_id = get_non_empty("\nEnter Item ID to update: ")

    item = find_item(item_id)

    if item is None:
        print("Item not found.")
        return

    print("\n========== UPDATE ITEM ==========")
    print("Press Enter to keep the current value.")

    new_name = input(
        f"Item name [{item[1]}]: "
    ).strip()
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if new_name != "":
    item[1] = new_name

new_category = input(
    f"Category [{item[2]}]: "
).strip()

if new_category != "":
    item[2] = new_category

while True:
    new_quantity = input(
        f"Quantity [{item[3]}]: "
    ).strip()

    if new_quantity == "":
        break

    try:
        new_quantity = int(new_quantity)

        if new_quantity > 0:
            item[3] = new_quantity
            break
        else:
            print("Quantity must be greater than 0.")

    except ValueError:
        print("Please enter a valid whole number.")

while True:
    new_condition = input(
        f"Condition [{item[4]}]: "
    ).strip()

    if new_condition == "":
        break

    new_condition = new_condition.capitalize()

    if new_condition in ["Good", "Damaged", "Missing"]:
        item[4] = new_condition

        if new_condition == "Good":
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            item[6] = "Available"
        elif new_condition == "Damaged":
            item[6] = "Damaged"
        else:
            item[6] = "Missing"

        break

    print("Please enter Good, Damaged, or Missing.")

new_location = input(
    f"Location [{item[5]}]: "
).strip()

if new_location != "":
    item[5] = new_location

print("\nItem updated successfully.")


def delete_item():
    """Delete an inventory item."""
    if len(inventory) == 0:
        print("\nNo inventory records to delete.")
        return

    display_items()

    item_id = get_non_empty(
        "\nEnter Item ID to delete: "
    )

    item = find_item(item_id)

    if item is None:
        print("Item not found.")
        return

    print(f"\nSelected item: {item[1]}")

    confirmation = input(
        "Are you sure you want to delete this item? (Y/N): "
    ).strip().lower()
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    if confirmation == "y":
        inventory.remove(item)
        print("Item deleted successfully.")
    else:
        print("Delete cancelled.")


def check_status():
    """Display damaged and missing inventory items."""
    if len(inventory) == 0:
        print("\nNo inventory records found.")
        return

    results = []

    for item in inventory:
        if (
            item[6].lower() == "damaged"
            or item[6].lower() == "missing"
        ):
            results.append(item)

    if len(results) > 0:
        print("\n========== DAMAGED / MISSING ITEMS ==========")
        display_items(results)
    else:
        print("\nNo damaged or missing items found.")


def show_summary():
    """Calculate and display inventory summary."""
    total_quantity = 0
    available_quantity = 0
    damaged_quantity = 0
    missing_quantity = 0

    for item in inventory:
        total_quantity += item[3]

        if item[6].lower() == "available":
            available_quantity += item[3]

        elif item[6].lower() == "damaged":
            damaged_quantity += item[3]
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        elif item[6].lower() == "missing":
            missing_quantity += item[3]

    print("\n========== INVENTORY SUMMARY ==========")
    print(f"Different item records : {len(inventory)}")
    print(f"Total quantity         : {total_quantity}")
    print(f"Available quantity     : {available_quantity}")
    print(f"Damaged quantity       : {damaged_quantity}")
    print(f"Missing quantity       : {missing_quantity}")


def show_menu():
    """Display the main menu."""
    print("\n========== LABTRACK ==========")
    print("1. Add Inventory Item")
    print("2. View Inventory")
    print("3. Search Inventory")
    print("4. Update Inventory")
    print("5. Delete Inventory")
    print("6. Check Damaged/Missing")
    print("7. Inventory Summary")
    print("8. Exit")
    print("==============================")


def main():
    """Run the LabTrack system."""
    while True:
        show_menu()

        choice = input(
            "Choose an option (1-8): "
        ).strip()

        if choice == "1":
            add_item()

        elif choice == "2":
            display_items()

        elif choice == "3":
            search_item()
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        elif choice == "4":
            update_item()

        elif choice == "5":
            delete_item()

        elif choice == "6":
            check_status()

        elif choice == "7":
            show_summary()

        elif choice == "8":
            print("\nThank you for using LabTrack!")
            break

        else:
            print("Invalid choice. Please select 1-8.")


main()
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